

1) Apriori Algorithm.

The Apriori Algorithm finds frequent item combinations by first finding frequent individual items and then generating larger items from them.

Minimum support count is 3.

Step 1: find frequent 1-itemsets

- mobile appears in 01, 02, 04, 05 $\rightarrow 4$
- earphones appears in 01, 03, 04, 05 $\rightarrow 4$
- charger appears in 02, 03, 04 $\rightarrow 3$

$$L_1 = \{\text{Mobile, earphones, charger}\}$$

Step 2: Generate 2-item candidates.

itemset	support count
Mobile - earphone	3
Mobile, charger	2
earphone, charger	2

After pruning itemsets with support less than 3

$$L_2 = \{\text{Mobile earphones}\}$$

Step 3: Generate 3-itemset

{mobile, Earphones, charger}

its support is 1, which is less than 3, so it is pruned

{mobile}, {Earphones}, {charger}, {Mobile, Earphone}

2) FP-Growth - library Book Borrowing.

FP-growth discovers frequent patterns without generating candidate itemsets.

It uses a compact structure called an FP-tree.

Minimum support count = 3

Step 1: Count individual books

- python = 4

- DataScience = 3

- machine learning = 4

Step 2: Arrange items according to frequency

python $\rightarrow$ Machine learning $\rightarrow$ DataScience.

Step 3: Construct the FP-Tree

The transactions are sorted using their order

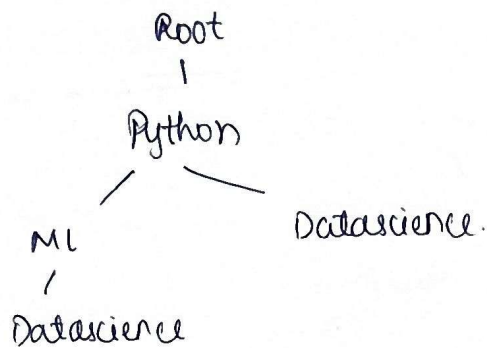
• B1 $\rightarrow$ python, DataScience.

• B2 $\rightarrow$ python, ML.

• B3 $\rightarrow$ ML, DataScience.

• B4 $\rightarrow$ python, ML, DS

• B5 $\rightarrow$ python, ML



Step 4: - Mine the FP-Tree

The tree is traversed to find frequent combination since FP-growth works directly on the compressed tree. Since FP-growth works directly on the compressed tree, it does not generate C_1, C_2, C_3 candidate sets like Apriori.

FP-growth is efficient for large datasets because it uses an FP-tree and avoids repeated candidate generation.

Q3 Association Rule Generation

Association rules are generated from frequent itemsets using confidence.

$$\text{confidence}(A \rightarrow B) = \frac{\text{support}(A \cup B)}{\text{support}(A)} \times 100$$

Assume minimum confidence = 60%

Rule 1

from $\rightarrow$ {Internet, Streaming} = 3

Internet $\rightarrow$ Streaming.

$$\text{confidence} = \frac{3}{6} \times 100 = 50\%$$

50% < 60%, the rule is accepted

Rule 2 streaming $\rightarrow$ Internet

$$\text{confidence} = 3/4 \times 100 = 75\%$$

Since $75\% > 60\%$, the rule is accepted.

Rule 2.

streaming $\rightarrow$ Internet

$$\text{confidence} = 3/4 \times 100 = 75\%$$

The rule is accepted.

Rule 3

streaming $\rightarrow$ cloud storage.

$$\text{confidence} = \frac{2}{4} \times 100 = 50\%$$

$50\% < 60\%$, the rule is rejected.

4. Constraint Based frequent itemset Mining

Every itemset must contain python

Minimum support count = 2

step 1: python frequency

python occurs in

- S1
- S2
- S4
- S5

$$\text{support}(\text{python}) = 4$$

step 2: Generate only python - containing combinations

Itemset	support
Python	4
python, Database	2
python, AI	2
python, Database, AI	1

The first three satisfy the minimum support. The 3-itemset

$\{\text{python, Database, AI}\}$

is removed because:

$$1 < 2.$$

$\{\text{python}\}$, $\{\text{python, Database}\}$, $\{\text{python, AI}\}$

effect of the constraint

The constraint prevents combinations such as:

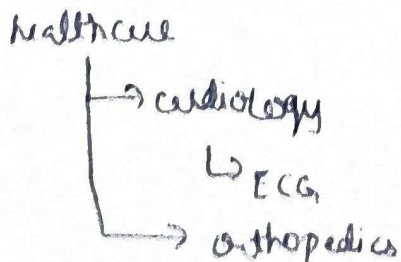
$\{\text{Database, AI}\}$

from being considered because they do not contain python

candidate generation becomes smaller and the mining process becomes more efficient.

5) Multilevel Association Rule Mining.

The treatment hierarchy is



Multilevel mining discovers relations at different levels of their hierarchy
minimum support count = 2

The treatment healthcare occurs in all five patients

$$\text{support}(\text{healthcare}) = 5$$

Cardiology occurs in:

P1, P3, P4

$$\text{support}(\text{cardiology}) = 3$$

orthopedics occurs in P3, P5

$$\text{support}(\text{orthopedics}) = 2$$

Possible rules are:

cardiology $\rightarrow$ healthcare

orthopedics $\rightarrow$ healthcare

Specific Treatment

ECA occurs in P1 & P4

$$\text{support}(\text{ECA}) = 2$$

ECA $\rightarrow$ cardiology

$$\text{confidence} = \frac{2}{2} \times 100 = 100\%$$

$\rightarrow$ level 1:

healthcare

$\rightarrow$ level 2:

cardiology $\rightarrow$ healthcare

orthopedics $\rightarrow$ healthcare

$\rightarrow$ level 3:

ECA $\rightarrow$ cardiology